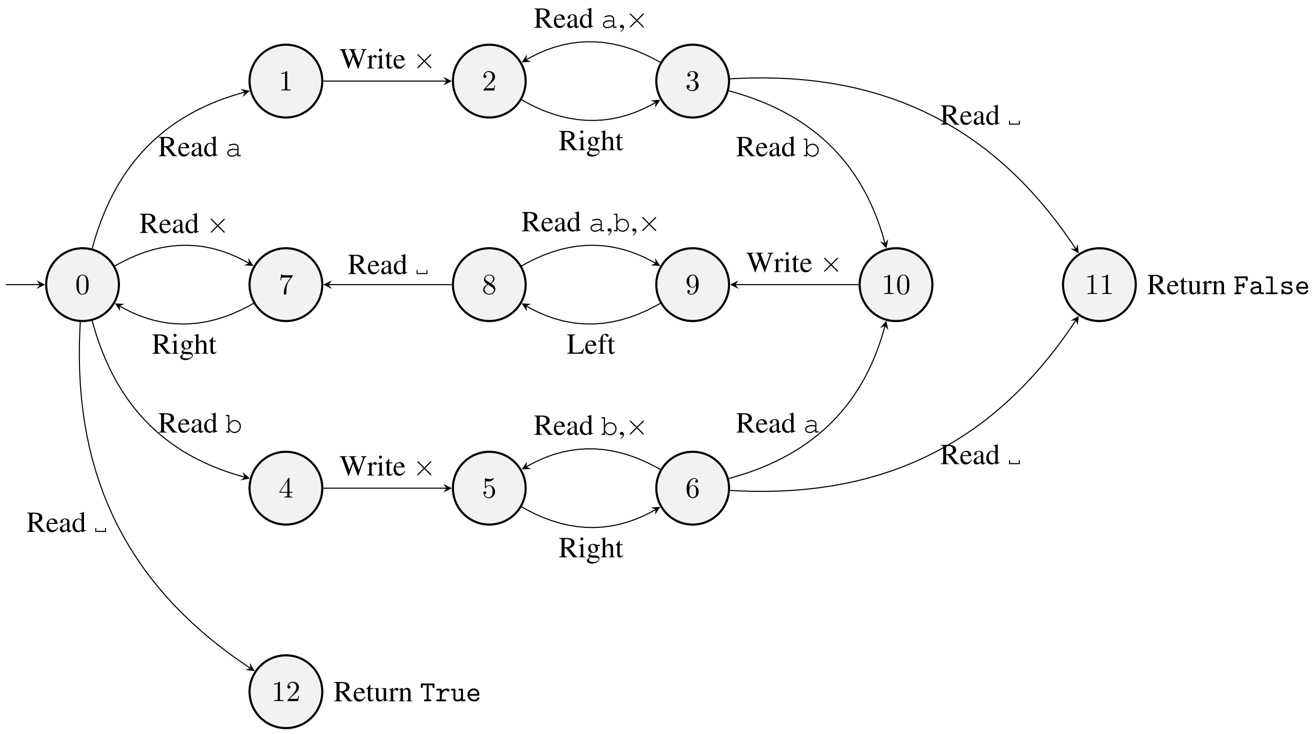
Theories of Computation: Summative Assignment 2

To be handed in on Canvas before **Tuesday 21st March, 12pm**

Consider the deterministic Turing machine $\mathcal{M}$ on alphabet $\{\texttt{a}, \texttt{b}, \times, \text{␣}\}$ with return value set $V = \{\texttt{True}, \texttt{False}\}$. The tape initially contains a non-empty block of a's and b's on an otherwise blank tape, with the head on the leftmost non-blank character. The transition function is given by the diagram below



Questions.

1. (a) Give the complete run of machine $\mathcal{M}$ on the word aa, with the head positioned on the first a.

   That is, the starting confiiguration of the tape is: ␣ $\overset{\bullet}{\texttt{a}}$ a ␣

   At each step, indicate the tape contents, the position of the head, the current state and the instruction (including the result if it is a Read). **[2 marks]**

   (b) Give the first seven steps of the run of machine $\mathcal{M}$ on the word ab, with the head positioned on the a.

   That is, the starting confiiguration of the tape is: ␣ $\overset{\bullet}{\texttt{a}}$ b ␣

   At each step, indicate the tape contents, the position of the head, the current state and the instruction (including the result if it is a Read). **[2 marks]**

2. (a) Is abaa in the language of $\mathcal{M}$? **[1 mark]**

   (b) Is abba in the language of $\mathcal{M}$? **[1 mark]**

   (c) What is the language of $\mathcal{M}$? **[2 marks]**

3. For $w \in \{\texttt{a}, \texttt{b}\}^*$, let $T(w)$ denote the processing time of machine $\mathcal{M}$ on word w.

   (a) Give a word w_n of length $n > 0$ such that $T(w_n)$ is linear in n. Justify your answer by giving $T(w_n)$ as a function of n and showing that it is in $\mathcal{O}(n)$. **[3 marks]**

   (b) Give a word v_n of length $n > 0$ (you may assume n is even) such that $T(v_n)$ is quadratic in n. Justify your answer by giving $T(v_n)$ as a function of n and showing that it is in $\mathcal{O}(n^2)$. **[3 marks]**

4. (a) Construct a deterministic Turing machine $\mathcal{N}$ on alphabet $\{\texttt{a}, \texttt{b}, \text{␣}\}$ with **seven** states that decides the language described by regular expression $\texttt{aa}^*\texttt{bb}^*$.

   Assume that the tape initially contains a non-empty block of a's and b's on an otherwise blank tape, with the head on the **rightmost** non-blank character.

   The position of the head at the end of the run does not matter. **[4 marks]**

   (b) Explain briefly how you would use machines $\mathcal{M}$ and $\mathcal{N}$ as macros to construct a Turing machine that recognizes language $\mathcal{L} = \{a^n b^n \mid n \geq 1\}$.

   That is, a machine that starts on a tape that initially contains a non-empty block of a's and b' on an otherwise blank tape (which represents a word $w \in \{\texttt{a}, \texttt{b}\}^*$) with the head on the **leftmost** non-blank character and returns `True` when $w \in \mathcal{L}$ and returns `False` when $w \notin \mathcal{L}$.

   The position of the head at the end of the run does not matter. **[2 marks]**

Solutions.

1. (a)

␣	$\overset{\bullet}{\texttt{a}}$	a	␣	0	Read a
␣	$\overset{\bullet}{\texttt{a}}$	a	␣	1	Write $\times$
␣	$\overset{\bullet}{\times}$	a	␣	2	Right
␣	$\times$	$\overset{\bullet}{\texttt{a}}$	␣	3	Read a
␣	$\times$	$\overset{\bullet}{\texttt{a}}$	␣	2	Right
␣	$\times$	a	$\overset{\bullet}{␣}$	3	Read ␣
␣	$\times$	a	$\overset{\bullet}{␣}$	11	Return `False`

   (b)

␣	$\overset{\bullet}{\texttt{a}}$	b	␣	0	Read a
␣	$\overset{\bullet}{\texttt{a}}$	b	␣	1	Write $\times$
␣	$\overset{\bullet}{\times}$	b	␣	2	Right
␣	$\times$	$\overset{\bullet}{\texttt{b}}$	␣	3	Read b
␣	$\times$	$\overset{\bullet}{\texttt{b}}$	␣	10	Write $\times$
␣	$\times$	$\overset{\bullet}{\times}$	␣	9	Left
␣	$\overset{\bullet}{\times}$	$\times$	␣	8	Read $\times$

2. (a) No, abaa is not in $L(\mathcal{M})$.

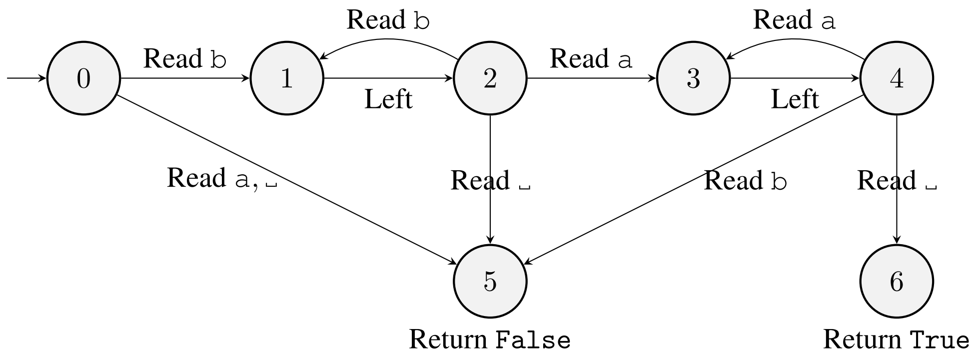
   (b) Yes, abba is in $L(\mathcal{M})$.

   (c) $L(\mathcal{M})$ is the language of words on $\Sigma = \{\texttt{a}, \texttt{b}\}$ that have the same number of a's and b's.

3. (a) Take $w_n = \mathtt{a}^n$.
Then, $T(w_n) = 2n + 3$
For $n \geq 1$, $\frac{T(w_n)}{n} \leq 5$, so $T(w_n)$ is in $\mathcal{O}(n)$.

(b) Assuming $n = 2p$, take $v_n = (\mathtt{ab})^p$.
Then, $T(v_n) = (\sum_{i=1}^{p}(6 + 2(2i-1) + 2 + 2(2i))) + 2 = 4p^2 + 10p + 2 = n^2 + 5n + 2$.
For $n \geq 1$, $\frac{T(v_n)}{n^2} \leq 8$, so $T(v_n)$ is in $\mathcal{O}(n^2)$.

4. (a)



(b) Start with the head on the leftmost character. Move all the way to the rightmost character. Run machine $\mathcal{N}$. If it returns `False`, then return `False`; otherwise, the head is on the blank immediately to the left of the word, so move one step to the right and run machine $\mathcal{M}$. If it returns `False`, then return `False`, otherwise return `True`.